MATH2: Week 6 Tutorial (Revision Session)

Q1. A sample of people have been assigned to fly to Mars. The frequencies of those in each age-group are given in the table below. Frequency represents the total number of people in each age range. Note, all ages are only measured (discretely) in years, and each range is inclusive of its boundaries.

(a) What is the Mode range for these data?

(b) What is the interpolated Median (Q) for these data?

(c) Calculate the Mean for these data.

Age range	Frequency
1 to 8	2
9 to 20	14
21 to 40	16
41 to 60	93
61 to 80	140
81 to 105	35

Q2. You are taking random samples of seashells on a beach. The seashells are either univalves or bivalves. You know that 76% of seashells on the beach are univalves. You pick 100 random samples of seashells from the beach. Taking each seashell is independent from the others and the probability of selecting a univalve or a bivalve remains the same for each sample.

Given that you hope to pick as many bivalves as possible, answer the following.

(a) What are the mean (expected value) and variance for taking 100 random samples of seashells from the beach?

(b) What is the probability that you select exactly one bivalve seashell in your 100 samples?

(c) What is the probability of collecting more than 86% univalve seashells in the 100 samples?

Q3. You have landed on an alien planet. When you go out to explore the planet you are guaranteed to encounter one alien. There are three types of aliens: Circles, Squares, and Triangles. You know that the chance of encountering a Circle is 50%, Square is 30% and Triangle is 20%.

Unfortunately, the aliens are sometimes unfriendly and may attack. This likelihood of being attacked depends on the type of alien you encounter. If you encounter (given) a Circle it has a 10% chance of attacking you, encountering (given) a Square has a 30% chance of attacking you, while encountering (given) a Triangle has an 80% chance of attacking you.

Given this information, answer the following questions.

(a) When you explore the planet, what is the probability of you being attacked by an alien?

(b) When you explored the planet: (given) if you were NOT attacked by an alien, what is the probability that you encountered a Circle?

(c) When you explored the planet: (given) if you were attacked by an alien, what is the probability that you encountered a Triangle?

Q4. Flying from earth to a distant planet may take a variety of different times, depending on random factors of space flight. The times are normally distributed with a mean time of 40 weeks and standard deviation of 5 weeks.

Given this information, answer the following.

(a) What is the probability that flying from earth to the planet takes <u>more than</u> 35 weeks?

(b) What is the probability that flying from earth to the planet takes <u>less than</u> 30 weeks?

(c) What is the probability that flying from earth to the planet takes <u>between</u> 25 and 45 weeks?

Q5. The average weight of otters is given as 7.1Kg and the standard deviation of this is given as 3Kg. The population weights are almost normally distributed. For a sample size of 80 otters, answer the following

(a) What is the mean and standard deviation of the sampling distribution of the mean (sample means for the sample size of 80)?

(b) What is the probability that a sample will have a mean of less than or equal to 7Kg (for the sampling distribution of the mean, of size 80 samples)?

(c) For the sampling distribution of the means (of size 80 samples), what is the range of means where 95% of the sample means are included? Write both the lowest mean value and the highest mean value (95% of the area of the distribution is between these two values).

Solutions:

Q1.

a) 61 to 80 (we are just using the basic Mode for this module)
b) 64.5714
c) 61.7383

Q2.

a) 24, 18.24
b) 3.8085e-11
c) 0.0069

Q3.

a) 0.3
b) 0.6429
c) 0.5333

Q4.

a) 0.8413
b) 0.0228
c) 0.84

Q5.

a) 7.1, 0.3354
b) 0.3821
c) 6.4426, 7.7574